16 An aqueous solution containing 0.10 mol dm^{-3} zinc sulfate and 0.10 mol dm^{-3} of copper(II) sulfate is saturated with hydrogen sulfide, H_2S, at 15 °C. The concentration of S^{2-}(aq) in the solution is 10^{-35} mol dm^{-3}.

The solubility product, K_{sp}, of zinc sulfide and copper(II) sulfide at 15 °C is given below.

	K_{sp} /mol^2 dm^{-6}
zinc sulfide	10^{-24}
copper(II) sulfide	10^{-40}

Which statement describes what happens in the solution?

A Only zinc sulfide is precipitated.

B Only copper(II) sulfide is precipitated.

C Copper(II) sulfide is precipitated, followed by zinc sulfide.

D More precipitate is observed by lowering the pH of the mixture.